

Multimodal Emotion Detection Using Deep Learning

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Abstract— Emotion recognition is a vital component of human communication and has significant applications across various domains. The goal of this research is to integrate text and visual data to create an effective model for real-time multimodal emotion identification. The text analysis utilizes fastText due to its speed and effectiveness, while 2D Convolutional Neural Network (CNN) is employed for image analysis to capture local features and handle content variations. The CMU-MOSEI dataset is used for testing and combining the datasets, and emotion predictions classified into six distinct categories. The study compares various approaches and emphasizes the advantages of multimodal affective computing systems, which achieve higher classification accuracy depending on factors like the number of emotions analyzed, extracted features, classification methods, and database consistency. By addressing challenges in understanding visual signals and emotional awareness, this research aims to advance emotion detection and recognition systems.

Keywords— CNN, fastText, Multimodal Emotion Detection

I. INTRODUCTION

Emotion recognition plays a vital role in enabling seamless interaction between humans and machines, as emotional cues are a crucial aspect of human communication. This technology has applications across various fields. For example, identifying emotional states can assist in monitoring and predicting fatigue levels. In speech recognition, it can be utilized in call centers to analyze the emotional state of callers and enhance service quality.

Recognizing emotions is a complex task because human emotions lack clear temporal boundaries, and individuals express and interpret emotions differently. While much of the focus in emotion recognition has been on analyzing speech to infer emotions, other modalities, such as visual cues (e.g., facial expressions), have also been explored.

The emergence of deep neural networks over the past decade has led to significant advancements in numerous established TF-IDF embeddings, reaching an F1-score of 92.18%. Ahuja et al. (2019) examined the effect of feature extraction on

pattern recognition domains, including object recognition, speech and speaker identification, and multimodal approaches like audio-visual recognition. These advancements have also made an impact in the growing field of paralinguistics.

The goal of this research is to investigate multimodal emotion detection and implement autonomous emotion identification methods in text and image domains that allow for quick and precise predictions.

The following are the study's objectives:

- To investigate multimodal emotion recognition and design architectures that are both accurate and efficient.
- To implement cross-dataset training and evaluation to build a resilient, general, and precise emotion recognition model.

II. RELATED WORK

Human emotion recognition has attracted a lot of research attention in recent years because of its numerous applications. This section presents the background of the subject and summarizes the literature. While multimodal emotion recognition is applicable to various fields, it still faces challenges that are gradually being addressed to improve model efficiency and practical usability.

A. Text emotion recognition

Several studies have advanced sentiment analysis using various techniques. Sailunaz and Alhaji (2019) used KNN and Naive Bayes models to achieve excellent accuracy on the ISEAR dataset. Liu (2020)[6] combined CBOW and CNN, achieving 87.2% and 90.5% accuracy on IMDB and COAE2014 datasets. Meng et al. (2019) used transfer learning with a multi-layer CNN for improved accuracy on Amazon and Chinese product reviews. Lee et al. (2021) employed a semi-supervised approach using a lasso-based ensemble model to handle limited data, achieving competitive results. Xu et al. (2019) enhanced BiLSTM predictions with sentiment-based sentiment analysis across machine learning and deep learning frameworks.

B. Image emotion recognition

Recent research has made significant advancements in sentiment and emotion recognition using image data. Babajee et al. (2020) combined CNN and SVM to predict seven emotion classes with 79.8% accuracy on the KDEP dataset. Pathak et al. (2020) and Yadav et al. (2020) used CNN-based models for facial emotion recognition, achieving 70% accuracy on a validation dataset and 57% on the FER-2013 dataset, respectively. Yadav and Vishwakarma (2020) introduced RA-DLNet for binary emotion classification, achieving 70% accuracy. Mutual training in a dynamic manner for semi-supervised learning was investigated by Feng et al. (2022), which may enhance generalization in spite of the scarcity of labeled data, albeit its efficacy in image emotion identification has not yet been proven. Zheng et al. (2020) highlighted the importance of local region features for visual emotion exploration, but they also pointed out possible issues with model efficiency and confidence ratings.

C. Multimodal emotion recognition

Recent progress in multimodal emotion recognition has investigated diverse architectures and methodologies. Kim and Lee (2020) proposed a Multi-Attention RNN Accuracy on the CMU-MOSI dataset was 84.31%. Sun et al. (2019) created multimodal embeddings that combined text and audio and they were able to classify sentiment with 93% accuracy. For the CMU-MOSI and CMU-MOSEI datasets, Yang et al. (2020) used Cross-Modal BERT with text-audio interaction to improve performance. Chaturvedi et al. (2019) fused ECG signals with text to predict 24 emotions, leading to a 10%–20% performance improvement. Yang et al. (2022) emphasized high-quality unimodal feature extraction, achieving 52.9% accuracy on CMU-MOSEI with a BERT-based approach. Shad Akhtar et al. (2019) introduced a multi-task, multi-attention framework using modality-specific transformers and cross-modal fusion, showing improvements over single-task models. Kumar and Vepa (2020) tackled multimodal challenges, achieving 81.1% accuracy on CMU-MOSEI by leveraging Bi-GRUs and cross-modal interactions. Guo et al. (2021) refined unimodal representations using LSTM and attention mechanisms to project features into a shared latent space. Mittal et al. (2021)[3] incorporated semantic and socio-dynamic interactions for context-aware analysis. Li et al. (2021) suggested a method inspired by quantum mechanics for modeling complex interactions. Huddar et al. (2021) used pairwise attention with RNNs for improved multimodal relationships, achieving 81.29% accuracy. Zhang et al. (2021) addressed data scarcity with To create a generalized framework that is impartial toward any particular dataset, several datasets were included for cross-

semi-supervised learning and cross-modal knowledge transfer, improving model generalization with unlabeled data.

D. Contributions

This study aims to contribute to multimodal emotion recognition by:

- Utilizing multiple benchmark datasets across individual modalities and their combined stacking, ensuring robust and generalized models capable of performing well on diverse datasets.
- Treating each modality independently, predicting emotions using separate models, and employing a lightweight stacking model for efficient real-time predictions.
- Making it possible to evaluate opinion films in real time for real-world applications using multiple categories and emotions.

III. MATERIALS AND METHODS

Section A provides an overview of the datasets utilized, describing their origins, characteristics, and role in this study. Section B provides the pre-processing done on the data obtained from these datasets for supporting training and evaluation of the respective models. Lastly, Section C provides a comprehensive description of the deep learning techniques and algorithms employed in developing the entire architecture. Figure 1 illustrates the entire workflow for training the text and image analysis model on its respective dataset.

A. Datasets

This research employs several datasets for various phases of emotion recognition, and there are three datasets in total: one for text emotion, one for image emotion, and one for multimodal testing and stacking.

For text-based emotion recognition, the "Emotions Dataset for NLP" from Kaggle was utilized. It comprises distinct training, validation, and testing subsets containing 16,000, 2,000, and 2,000 samples, respectively. It is imbalanced, with happiness having the utmost number of entries and surprise the least across all subsets. Table 1 presents a summary of the class-wise distribution for each emotion across each subset.

For visual emotion recognition, the "FER-2013" dataset (Goodfellow et al., 2013)[5] from Kaggle was employed. It consists of 35,889 samples spanning seven emotion categories. Table 2 outlines the class-wise distribution, detailing the number of images corresponding to each emotion in each subgroup. The images are grayscale, measuring 48x48 pixels.

dataset training and assessment. Every dataset depicts human emotions that are frequently expressed, despite minor

differences in class labeling. These differences are standardized in the final multimodal estimate to focus on the six most prevalent emotions: happiness, sadness, anger, surprise,fear,disgust.

TABLE I: CLASS-WISE DISTRIBUTION OF TEXT DATABASE

Subset	Sadness	Anger	Love	Surprise	Fear	Happiness	Total
Training	4666	2159	1304	572	1937	5362	16000
Validation	550	275	178	81	212	704	2000
Testing	581	275	159	66	224	695	2000
Total	5797	2709	1641	719	2373	6761	20000

TABLE II: CLASS-WISE DISTRIBUTION OF IMAGE DATASET

Subset	Anger	Disgust	Fear	Happiness	Neutral	Sadness	Surprise	Total
Train	3995	436	4097	7215	4965	4830	3172	28710
Test	958	111	1024	1774	1233	1247	832	7179
Total	4953	547	5121	8989	6077	6077	4004	35889

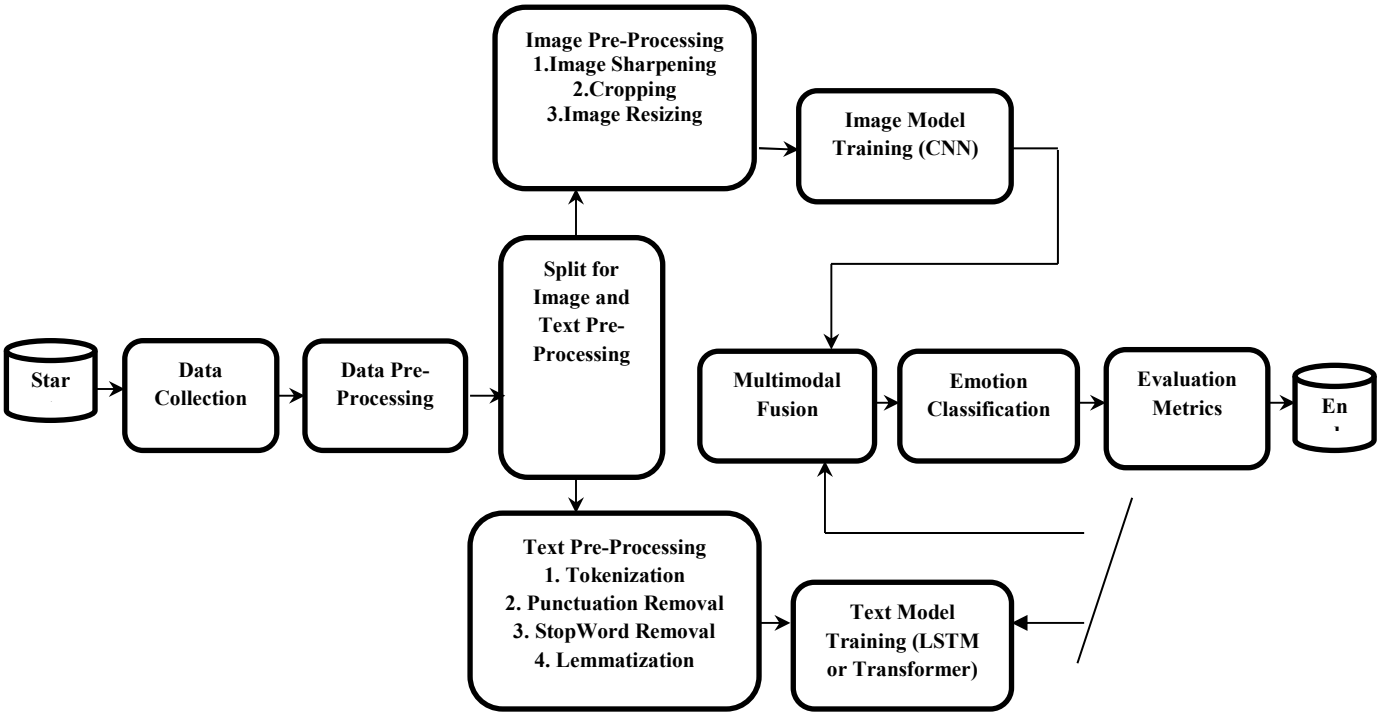


Fig.1. Block diagram for Text and Image Analysis

B. Data Preprocessing

For text-based classification, each sentence in the transcript is an independent observation to classify the respective

sentiment. Sentence-level segmentation preserves the context of individual words, enhancing understanding and prediction. For real-time processing, the model does not finalize the prediction until a sentence is finished. Each sentence undergoes the following steps: punctuation removal, conversion to lowercase, tokenizing, stop word removal, and lemmatization. Refined text with few but essential keywords is now prepared to be input into the text emotion recognition model.

For image-based emotion identification, each image in the dataset is pre-processed using the Image Data Generator to enhance and expand the dataset through augmentation. The data is augmented with variations such as angle adjustments, stretching, and zooming while preserving the target resolution of 48x48 pixels. This upgraded dataset is then readied to be fed into the 2D CNN model for classifying emotions.

The pre-trained DeepFace module for Python is used to handle the frames collected from the multimodal dataset at one-second intervals, with OpenCV acting as the face recognition backend. If a face is identified, then the face is clipped and the frame is turned to grayscale. A 3x3 sharpening filter is then applied to improve the image:

$$K = \begin{bmatrix} 0 & -1 & 0 \\ -1 & 5 & -1 \\ 0 & -1 & 0 \end{bmatrix}$$

$$-1 \ 5 \ -1$$

Finally, the image is resized to 48x48 pixels before being input into the model for prediction.

C. Methodology

The overall framework is structured into three separate modules. The first two modules correspond to the individual components of multimodal input, each processing text and image data independently. The final module is dedicated to real-time testing through a graphical user interface (GUI), allowing users to interact with the system and view predictions based on the processed multimodal inputs.

D. Text emotion recognition

The dataset was used to train the text emotion identification model, which used the fastText model for sentiment classification. fastText, developed by Facebook Research, was employed to improve the precision and efficacy of emotion detection., uses a shallow neural network with a continuous bag of words and character n-grams, making it efficient for cross-corpus tasks and handling rare words. For multi-class classification, the model was trained across eight epochs using a one-vs-all loss function and a learning rate of 0.7. This approach excels in real-time analysis, especially with unknown words, due to its ability to handle imbalanced classes effectively. For multimodal sentiment prediction, sentences from the dataset were pre-processed, and

confidence scores were aggregated to rank the predicted sentiments, with a minimum threshold set to include all sentiment possibilities.

E. Image emotion recognition

For image-based emotion identification, the FER-2013 dataset from Kaggle was utilized. This dataset comprises grayscale facial images depicting various expressions. To expand and enhance the dataset, the Image Data Generator module from TensorFlow was employed to generate augmented image distinctions.

The processed data was then fed into a 2D CNN, a widely used model for image identification and object detection tasks. Two convolutional layers, two batch normalization layers, a dropout layer, a max-pooling layer, and a flattened layer were among the four components that made up the CNN architecture. Batch normalization, dropout layers, and dense layers were incorporated in the next blocks. A final dense layer produced sentiment probability outputs.

The model was trained over 133 epochs, producing sentiment predictions as probability vectors. For opinion videos, pre-processing involved face detection and image refinement to isolate the facial region, with sentiment confidence scores averaged to determine the dominant emotions present in the video. The training steps were illustrated in a flow diagram to ensure consistency between the training and testing datasets.

F. Model testing, aggregation and GUI

The third section is utilized after model training, where the trained models are applied to the CMU-MOSEI dataset, which includes video transcripts. For the videos, any feeling that is not zero is deemed legitimate. Each model's pre-processing, modality extraction, and predictions are carried out separately, as previously mentioned. Each model's outputs are then combined using a Late-fusion Random Forest model; stacking is the recommended method because of its better performance. The model input consists of an 18-element vector, where each element is either 1 (indicating the presence of a sentiment) or 0 (indicating its absence). The output is a 6-element vector representing the six sentiment categories: anger, disgust, fear, happiness, sadness, and surprise. The Multi Output Classifier from the sklearn library is used to handle the multi-output nature of the problem.

After training and optimizing the models on the multimodal dataset, Tkinter was used to create a simple graphical user interface for real-time video analysis. The GUI reduces latency by introducing minimal delays. The text model

pauses until a sentence is fully completed, the audio model processes each audio segment with a 3-second delay, and the image model captures frames at 1-second intervals. The same approach is applied to the CMU-MOSEI dataset for real-time evaluation, where the individual model outputs are continuously integrated into the stacking model to refine the overall sentiment predictions.

IV. RESULT AND DISCUSSION

All models achieved accuracy levels comparable to modern techniques. Even after being assessed using the CMU-MOSEI dataset, which was not used during training, the architecture demonstrated strong performance.

Figures 3(a) and 3(b) illustrate the relationship between epochs and accuracy for both training and validation across all models. The orange lines denote testing accuracy, while the blue lines indicate training accuracy. The number of epochs and details regarding the deep learning model layers are outlined in Section C. From the graphs, it can be observed that in Figure 3(a), accuracy rises sharply during the first two epochs, followed by a gradual improvement. The gap between training and evaluation accuracy was narrower for alternative loss functions, but the selected function was specifically designed for multi-class classification, ensuring a balance between generalization and accuracy.

The intricacy of the fastText model is responsible for the graph's smoothness, while the other models were created separately. Figure 3(b) for the image model reveals an unusual pattern, where testing accuracy continues to increase until approximately 15 epochs, after which fluctuations indicate varying improvements. Figures 2(a) and 2(b) present the confusion matrices for each model.

The datasets for each model were split in an 80-20 ratio for training and evaluation. The models achieved the best accuracy on the training dataset after fine-tuning; nevertheless, to maximize training and testing performance, the loss function needed to be adjusted. As a result, the ultimate accuracy was 81.56%, which was a trade-off between somewhat lower accuracy and better generality.

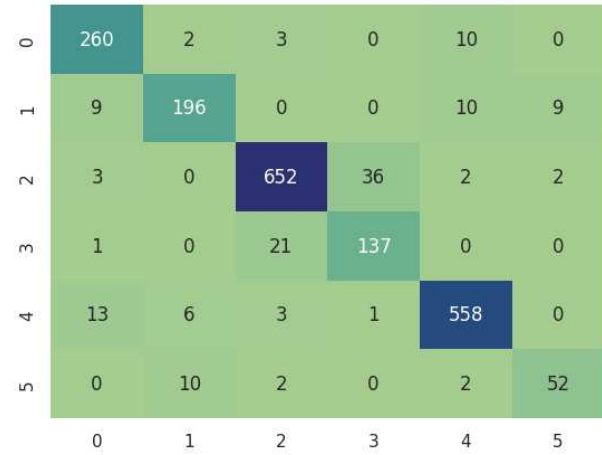


Fig.2(a). Confusion matrix for Text analysis

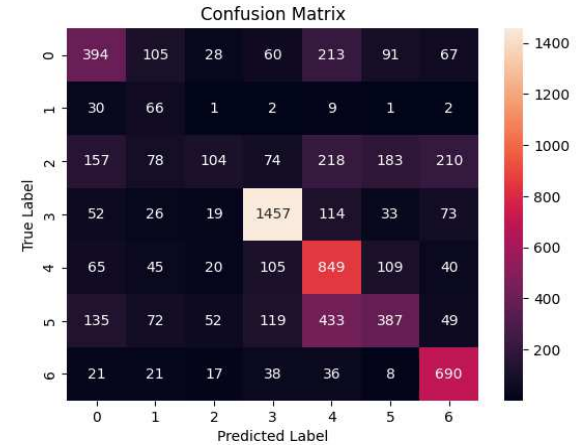


Fig.2(b). Confusion matrix for Emotion analysis

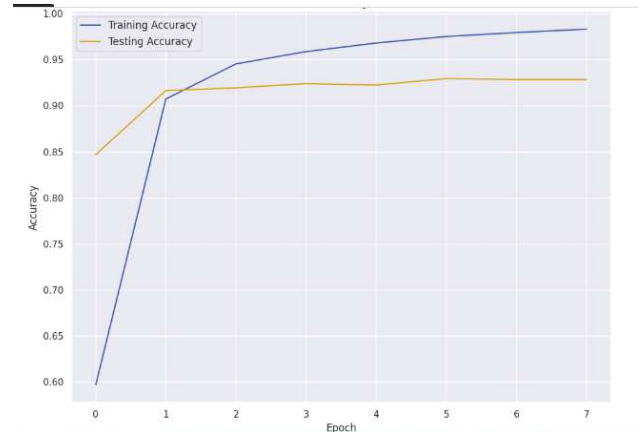


Fig.3 (a) Text Emotion Recognition

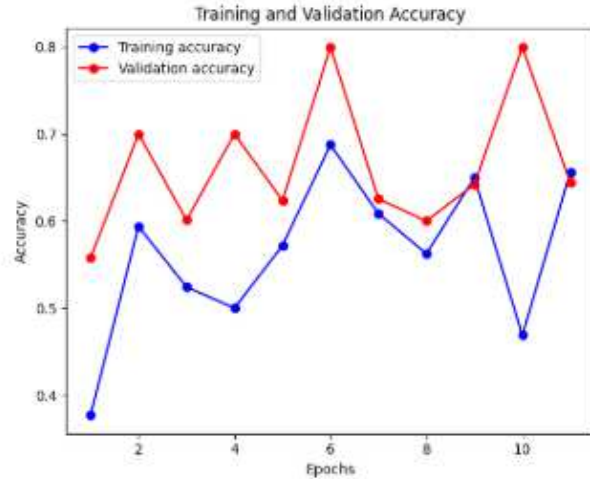


Fig.3 (b) Image Emotion Recognition

Fig.3. Epoch vs accuracy for each model

V. APPLICATIONS IN REAL TIME

The proposed multimodal emotion identification system has great potential for use in many areas due to its real-time analysis. This makes it more practical than other models. In healthcare, it can help monitor patients' behavior, aiding mental health diagnoses and guiding doctors in treatment (Haider et al., 2021).[4]

In law enforcement, it can analyze criminal behavior, helping to detect patterns and predict reoffending [6](Liu & Liu, 2021). In politics, it can assess public reactions to speeches, helping speakers improve their addresses. Before elections, it can reveal how speeches influence people and detect incitement or manipulation (Zhao et al., 2019).

It can also fight misinformation by tracking harmful emotional cues in political statements or social media (Ogundokun et al., 2022)[1]. In business, it improves customer feedback analysis, helping companies understand customer feelings and improve their offerings (Yu et al., 2022).[6]

Though designed for emotion recognition, its training approach can be adapted for broader prediction tasks beyond emotions.

Discard frames where no face is detected. However, the model has not been specifically tested for scenarios with multiple faces present in the frame.

VI. CONCLUSION AND FUTURE SCOPE

This research introduces a cross-dataset framework for concurrent evaluation of opinion films by integrating text and visual data. Separate models are trained on individual datasets and combined using a Random Forest stacking method to reduce biases and enhance accuracy. The architecture achieved 81.56% accuracy and a 75 F1-score on the CMU-MOSEI dataset, performing competitively with state-of-the-art models. Its fast design makes it ideal for real-time use. Key challenges include boosting accuracy, using pre-trained models, and addressing the limited representation of "disgust." Expanding to various languages and ethnicities can improve its adaptability and robustness.

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